

7/22/23, 2:21 AM

Codio - Manipulating Images

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SANDIPAN-DEY

exerc.pyx

```
1 import os
2 import openai
3 import secret
4 from PIL import Image
5 import io
6 from io import BytesIO
7 import requests
8
9 openai.api_key='sk-QVkkVGvk7ose2G3HCQM7T3B1bkFJI8YuDKIMdS8Rk6rTsgiy'
10 # WRITE YOUR CODE HERE
11
12 def generate_image(prompt):
13     response = openai.Image.create(
14         prompt=prompt,
15         n=1,
16         size="512x512"
17     )
18     image_url = response['data'][0]['url']
19     img_data = requests.get(image_url).content
20     with open('test_img.png', 'wb') as handler:
21         handler.write(img_data)
22     image = Image.open('test_img.png')
23     image = image.convert('L')
24     image = image.resize((256, 256), Image.ANTIALIAS)
25     image.save('test_img.png')
26
27
```

71% (19:37)

Terminalx

```
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 5.19.0-1028-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

 * Canonical Livepatch is available for installation.
   - Reduce system reboots and improve kernel security. Activate at:
```

Guidex

Collapse

Coding Exercise

Objectives

1. Use the openai.Completion.create method to generate an image from DALL-E.

2. Convert the image to grayscale using PIL.

3. Resize the image to 256x256 pixels using PIL.

4. Save the manipulated image to the local file system.

Your task is to create a Python script that generates a DALL-E image based on the prompt and then applies a series of transformations to it using PIL.

The prompt should be a function argument in the function `generate_image(prompt)`.

The transformations to apply include:

1. Converting the image to grayscale

2. Resizing the image to 256x256 pixels

3. After applying these transformations,your script should save the image to the local file system. it should be saved under `test_img.png`

Your script should use `openai.Completion.create` to generate a DALL-E response, and the PIL library to manipulate the resulting image.

Grayscale

Here is a simple technique to gray scale your image.
Make sure to replace 'your_image.jpg' with the path to your actual image file.

Convert the image to grayscale using the `convert()` method with the 'L' mode:

```
gray_image = image.convert('L')
```

Optionally, you can save the grayscale image using the `save()` method:

```
gray_image.save('grayscale_image.jpg')
```

Replace 'grayscale_image.jpg' with the desired filename and extension.

Putting it all together, here's an example:

```
from PIL import Image

# Open the image
image = Image.open('your_image.jpg')

# Convert to grayscale
gray_image = image.convert('L')

# Save the grayscale image
gray_image.save('grayscale_image.jpg')
```

After running this code, you will have a grayscale version of your image saved as `grayscale_image.jpg`.

TRY IT!

Command was successfully executed.

Click here to refresh your test image

When ready to submit, please click on the button below.

```
def generate_image(prompt):
    response = openai.Image.create(
        prompt=prompt,
        n=1,
        size="512x512"
    )
    image_url=response['data'][0]['url']

    # Download the image
    response = requests.get(image_url)
    img = Image.open(io.BytesIO(response.content))
    # Apply transformations
    img = img.convert("L") # Convert to grayscale
    img = img.resize((256, 256)) # Resize to 256x256 pixels
    # Save the image
    img.save("test_img.png")
```

Check IT!

LAST RUN on 7/22/2023, 2:20:47 AM

Check 1 passed

Mark as Completed

Back to dashboard

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